1 Introduction

Rainwater harvesting means collecting rain from a roof and storing it for later use. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

1.1 Why harvest rainwater

It reduces dependence on municipal supply and lowers groundwater extraction. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

In dry months a full tank can supply toilets and gardens for several weeks. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

1.2 Legal status

Under the Water Act 2019, households may use harvested rainwater for up to 10% of their drinking supply after certified filtration. Non-potable use has no limit. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

2 System Components

Every system has four parts: catchment, conveyance, first-flush diversion and storage. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

2.1 Catchment

Metal roofs give the cleanest runoff, with a runoff coefficient near 0.9. Clay tiles reach about 0.8 and thatch only 0.3. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

2.2 First-flush diverter

The first-flush diverter discards the first 2 litres per square metre of roof, which carries dust, leaves and bird droppings. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

2.3 Filters

A 200 micron mesh filter protects the tank inlet; a sand filter is needed for drinking use. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

3 Design

Design starts from how much water falls on the roof and how much the household needs. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

3.1 Storage tank sizing

Tank capacity in litres equals roof area in square metres times annual rainfall in millimetres times the runoff coefficient. For a 50 square metre metal roof and 800 mm of rain this gives 36,000 litres per year. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

3.2 Maintenance schedule

Clean gutters monthly in the rainy season, flush the diverter after every storm and disinfect the tank once a year with chlorine at 5 mg per litre. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.

4 Conclusion

A well designed system is cheap, simple and pays for itself. Collected water should be inspected regularly for colour and smell. Households that plan the system carefully usually recover the installation cost within a few seasons. Local climate, roof material and family size all change the right design, so each choice below explains the trade-offs involved.